

3.6 Chain Rule Worksheet

1. Using Leibniz notation $\left(i.e., \frac{dy}{dx}, \frac{dP}{dt}, etc.\right)$, find the derivatives for each of the following functions.

a) $w = 10(q+1)^8 e^q + e^{\cos\left(\frac{\pi}{12}\right)}$

b) $x = 10y^3 \sin(2y) + \ln\left(\frac{5}{e^2}\right)$

c) $q = \frac{4 \sin(t)}{t^2 + 3}$

d) $k = \sec(3t) + \sin\left(\frac{1}{e}\right)$

e) $y = \frac{(3x^2 - 1)^2}{\sin(e^x)}$

f) $g = \cos(\tan(x^8))$

g) $v = e^{13t} \cot(25t)$

h) $w = \sin\left(\frac{v^{12}}{e^v}\right)$

2. Let $h(x) = \sqrt{4 + 3f(x)}$. If $f(1) = 7$ & $f'(1) = 4$. Find $h'(1)$.

3. Differentiate $y = \cot(x^4 + 8x)$.

4. Differentiate $y = [4 \sin(x) + 2 \cos(x)]^{3/2}$.

5. Differentiate $y = \sqrt{\sin(e^{t^3+4t})}$.

6. Differentiate $y = \left[\frac{x^2}{x^5 + 1} \right]^4$.

7. Differentiate $y = \sin(\cos(\tan(x)))$.

8. Determine the equation of the line which is tangent to the curve $y = 4 \tan\left(\frac{x}{4}\right) + 5 \cos(x) + \pi$ at $x = \pi$.

9. One of the points on the graph of $f(x) = e^{-9x}$ has the property that the line tangent to the curve at that point is perpendicular to the line $y = 9x + 3$. Find and fully simplify the y -coordinate for this point.

10. Find the equation of the line which is tangent to the curve $y = 2e^x - 6$ at its x -intercept.